

3) b) derive relationship of x and y and the slope and intercept of a log-log plot

for a straight line:

$$y = mx + b$$

A log-log plot plots the $\log(x)$ against $\log(y)$, so this turns into

$$\log(y) = m \log(x) + b \rightarrow 10^{\log(y)} = 10^{m \log(x) + b} \rightarrow y = 10^{\log x \cdot m} \cdot 10^b$$

$$\rightarrow y = x^m 10^b$$

A straight line on a log-log plot corresponds to a power law on an x - y plot.

$$\log(y) = m \log(x) + b$$

$$y = x^m 10^b$$